Siddharth Saminathan

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Chennai, India

PROFILE

AI PRODUCT ENGINEER WITH 4+ YEARS BUILDING AND SHIPPING PRODUCTION AI SYSTEMS THAT AUTOMATE MISSION-CRITICAL ENTERPRISE WORKFLOWS, FROM AI-DRIVEN QUALITY MANAGEMENT (DFMEA. PFMEA, CONTROL PLANS, 8D) TO REAL-TIME AI PRODUCTS AT SCALE. I TAKE SYSTEMS FROM CONCEPT TO PRODUCTION AND OPTIMIZE THEM RELENTLESSLY: LATENCY, COST, RELIABILITY, AND USER OUTCOMES. MULTI-AGENT ORCHESTRATION, TOOL-DRIVEN ARCHITECTURES, AND MEMORY SYSTEMS. WHAT I BUILD DOESN'T JUST RESPOND, IT IMPROVES OVER TIME.

{{SUMMARY_TEXT}}

{{SECTION_COMPETENCIES}}

{{COMPETENCIES}}

WORK EXPERIENCE

AI ENGINEER — OMNEX SYSTEMS, CHENNAI
JUL 2025 - PRESENT

OMNEX IS AN ENTERPRISE AI QUALITY PLATFORM (EQMS) SERVING GLOBAL MANUFACTURING. I BUILT THEIR AGENTIC AI SYSTEM FOR END-TO-END MANUFACTURING QUALITY DIGITALIZATION, AUTOMATING THE CORE QUALITY ENGINEERING WORKFLOWS THAT AUTOMOTIVE, AEROSPACE, AND INDUSTRIAL OEMS RUN ON.

- BUILT A PRODUCTION MULTI-AGENT AI SYSTEM THAT AUTOMATES END-TO-END MANUFACTURING QUALITY WORKFLOWS: DFMEA (DESIGN FAILURE MODE), PFMEA (PROCESS FAILURE MODE), CONTROL PLANS, AND 8D CORRECTIVE ACTION, REDUCING MANUAL ENGINEERING EFFORT AND IMPROVING CONSISTENCY ACROSS GLOBAL SUPPLY CHAINS.
- DESIGNED A TOOL-DRIVEN ORCHESTRATION LAYER THAT DYNAMICALLY SELECTS TOOLS, ROUTES DATA ACROSS INTERNAL AND EXTERNAL ENTERPRISE SYSTEMS, AND EXECUTES MULTI-STEP QUALITY WORKFLOWS (GENERATE TO VALIDATE TO UPDATE TO ACT) WITH FULL TRACEABILITY.
- ENABLED CONTEXT-AWARE QUALITY DECISIONING: THE SYSTEM MAPS INPUTS TO PRODUCTION HIERARCHIES (PART/PROCESS/PLANT), RETRIEVES RELEVANT HISTORICAL QUALITY CASES, AND SUGGESTS STRUCTURED CORRECTIVE ACTIONS AND PROCESS UPDATES.
- IMPROVED SYSTEM PERFORMANCE AND RELIABILITY, REDUCED WORKFLOW LATENCY (~12.5S TO ~8S), BUILT VALIDATION LAYERS AND RETRY LOGIC, AND INSTRUMENTED FULL OBSERVABILITY (LOGGING, TRACING, METRICS) FOR PRODUCTION MANUFACTURING ENVIRONMENTS.

CO-FOUNDER / AI ENGINEER

EMOTE (EMOTENOW.APP)
APR 2024 - PRESENT

- BUILT AND ITERATED A PRODUCTION AI SYSTEM THAT LEARNS FROM USER INTERACTIONS OVER TIME, MODELING CONTEXT, RELATIONSHIPS, AND BEHAVIORAL PATTERNS ACROSS SESSIONS TO IMPROVE RESPONSE QUALITY AND CONTINUITY
- DESIGNED AND SCALED THE END-TO-END BACKEND ARCHITECTURE (FASTAPI, REDIS, POSTGRESQL, REAL-TIME PIPELINES, MULTI-MODEL ROUTING), ENABLING LOW-LATENCY, STATEFUL AI INTERACTIONS IN PRODUCTION
- DROVE CORE SYSTEM PERFORMANCE AND EFFICIENCY, REDUCING LATENCY (~15S TO ~3S) AND INFERENCE COST (~$1.20 TO ~$0.023) THROUGH PIPELINE OPTIMIZATION, CACHING, AND MODEL SELECTION
- OWNED THE FULL PRODUCT LOOP (0 TO 1 TO ITERATION): SCALED TO 450+ USERS WITH 40+ POWER USERS, IMPROVED ACTIVATION (20% TO 75%) AND RETENTION (5% TO 13-15%) BY CONTINUOUSLY SHIPPING IMPROVEMENTS BASED ON REAL USER BEHAVIOR AND FEEDBACK.

DATA ENGINEER

AKKODIS AB, SWEDEN
NOV 2022 - NOV 2024

- LED MIGRATION FROM AZURE TO POSTGRESQL, AUTOMATING ETL PROCESSES USING DOCKER, ARGO, AND PYTHON.
- DEVELOPED AND OPTIMIZED CRON JOBS TO ENHANCE WORKFLOW EFFICIENCY.
- IMPROVED POWER BI DASHBOARDS AND METRICS PIPELINES, RESULTING IN 5% INCREASE IN BUSINESS REPORTING ACCURACY.
- CONSTRUCTED AUTOMATED VALIDATION LAYERS TO ENSURE DATA QUALITY AND INTEGRITY ACROSS ENTERPRISE REPORTING SYSTEMS.
- ENHANCED ETL WORKFLOWS THROUGH DOCKER CONTAINERIZATION AND POSTGRESQL OPTIMIZATION.

DATA ANALYST

POSITIVE INTEGERS PVT LTD, CHENNAI
SEP 2019 - MAY 2020

- ANALYZED FINANCIAL DATASETS RELATED TO LOAN PORTFOLIOS, CREDIT RISK, AND CUSTOMER BEHAVIOR PATTERNS FOR AN NBFC
- DEVELOPED INTERACTIVE DASHBOARDS USING TABLEAU AND GRAFANA TO VISUALIZE BUSINESS KPIS AND FINANCIAL METRICS
- CONSTRUCTED SQL-BASED ETL WORKFLOWS.

{{EXPERIENCE}}

{{SECTION_PROJECTS}}

{{PROJECTS}}

EDUCATION

M.SC. STATISTICS AND MACHINE LEARNING - LINKÖPING UNIVERSITY, SWEDEN (2020-2022)

B.TECH COMPUTER SCIENCE AND ENGINEERING - SRM UNIVERSITY, INDIA (2016-2020)

MASTER'S THESIS

DEVELOPED A MACHINE LEARNING PIPELINE UTILIZING AUTOENCODERS AND CLASSIFIERS TO PREDICT COLORECTAL CANCER METASTASIS FROM RNA SEQUENCING DATA. EMPLOYED TRANSFER LEARNING, PCA, AND NETWORK ANALYSIS FOR BIOMARKER DETECTION IN HIGH-DIMENSIONAL BIOLOGICAL DATASETS.

{{EDUCATION}}

{{SECTION_CERTIFICATIONS}}

{{CERTIFICATIONS}}

SKILLS

PROGRAMMING	**PYTHON · SQL**
AI SYSTEMS	**LLM APIS · RAG PIPELINES · EMBEDDINGS · PROMPT ENGINEERING · TOOL CALLING · MULTI-MODEL ROUTING · AGENTIC WORKFLOWS · LANGCHAIN/LANGGRAPH**
BACKEND	**FASTAPI · REDIS (CACHING, QUEUES) · POSTGRESQL · REST APIS · ASYNC PIPELINES · WEBSOCKETS. SSE STREAMING**
SYSTEMS DESIGN	**STATEFUL SYSTEMS · WORKFLOW ORCHESTRATION · MEMORY SYSTEMS · CONTEXT MODELING · DISTRIBUTED PIPELINES**
INFRA	**DOCKER · AWS · FLY.IO · LOGGING · TRACING · METRICS · PERFORMANCE OPTIMIZATION**
DATA	**ETL PIPELINES · DATA VALIDATION · POWER BI · TABLEAU**

{{SKILLS}}